

Docker-compose

1. Create and change directory, install docker-compose

```
# dnf install docker-compose-plugin -y
```

```
# mkdir docker-compose
```

```
# cd docker-compose
```

```
[root@rhel9 ~]# dnf install docker-compose-plugin -y
Updating Subscription Management repositories.
Last metadata expiration check: 0:03:06 ago on Thu 23 Jul 2026 02:24:03 PM +04.
Package docker-compose-plugin-5.3.1-1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@rhel9 ~]# docker compose version
Docker Compose version v5.3.1
[root@rhel9 ~]#
```

```
[root@rhel9 ~]# mkdir docker-compose
[root@rhel9 ~]# cd docker-compose/
[root@rhel9 docker-compose]# pwd
/root/docker-compose
[root@rhel9 docker-compose]#
```

2. Create docker-compose file

```
# vim docker-compose.yml
```

```
version: '3.9'
```

```
services:
```

```
  web:
```

```
    build: .
```

```
    ports:
```

```
      - "8000:5000"
```

```
    depends_on:
```

```
      - redis
```

```
  redis:
```

```
    image: redis:alpine
```

```
version: '3.9'

services:
  web:
    build: .
    ports:
      - "8000:5000"
    depends_on:
      - redis

  redis:
    image: redis:alpine
```

3. Create Dockerfile

vim Dockerfile

```
FROM python:3.7-alpine
WORKDIR /code
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
RUN apk add --no-cache gcc musl-dev linux-headers
COPY requirements.txt requirements.txt
RUN pip install -r requirements.txt
EXPOSE 5000
COPY . .
CMD ["flask", "run"]
```

```
FROM python:3.7-alpine
WORKDIR /code
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
RUN apk add --no-cache gcc musl-dev linux-headers
COPY requirements.txt requirements.txt
RUN pip install -r requirements.txt
EXPOSE 5000
COPY . .
CMD ["flask", "run"]
```

4. Create app.py file

vim app.py

```
import time
import redis
from flask import Flask

app = Flask(__name__)

cache = redis.Redis(host='redis', port=6379)

def get_hit_count():
    retries = 5
    while True:
        try:
            return cache.incr('hits')
        except redis.exceptions.ConnectionError as exc:
            if retries == 0:
                raise exc
            retries -= 1
            time.sleep(0.5)
```

```

@app.route('/')
def hello():
    count = get_hit_count()
    return 'Hello World! I have been seen {} times.\n'.format(count)

```

```

import time
import redis
from flask import Flask

app = Flask(__name__)

cache = redis.Redis(host='redis', port=6379)

def get_hit_count():
    retries = 5
    while True:
        try:
            return cache.incr('hits')
        except redis.exceptions.ConnectionError as exc:
            if retries == 0:
                raise exc
            retries -= 1
            time.sleep(0.5)

@app.route('/')
def hello():
    count = get_hit_count()
    return 'Hello World! I have been seen {} times.\n'.format(count)

```

5. Create requirements.txt file

```
# echo -e "flask\nredis" > requirements.txt
```

```

[root@rhel9 docker-compose]# echo -e "flask\nredis" > requirements.txt
[root@rhel9 docker-compose]# cat requirements.txt
flask
redis
[root@rhel9 docker-compose]# █

```

```

[root@rhel9 docker-compose]# pwd
/root/docker-compose
[root@rhel9 docker-compose]# ll
total 16
-rw-r--r--. 1 root root 434 Jul 23 14:52 app.py
-rw-r--r--. 1 root root 180 Jul 23 14:45 docker-compose.yml
-rw-r--r--. 1 root root 259 Jul 23 14:47 Dockerfile
-rw-r--r--. 1 root root 12 Jul 23 15:02 requirements.txt
[root@rhel9 docker-compose]# █

```

6. Run project

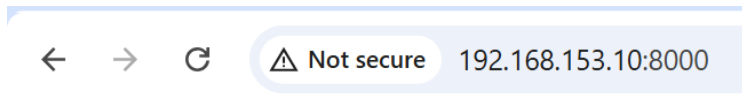
docker compose up -d --build

```
[root@rhel9 docker-compose]# docker compose up -d --build
WARN[0000] /root/docker-compose/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 8/10
Image redis:alpine [#####] 35.65MB / 37.43MB Pulling 42.5s
[+] Building 118.8s (13/13) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 494B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 358B 0.0s
=> [internal] load metadata for docker.io/library/python:3.7-alpine 3.5s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/6] FROM docker.io/library/python:3.7-alpine@sha256:f3d31c8677d03f0b3c724446077f229a6ce9d3ac430f5c08cd7dff00292048c3 11.9s
=> => resolve docker.io/library/python:3.7-alpine@sha256:f3d31c8677d03f0b3c724446077f229a6ce9d3ac430f5c08cd7dff00292048c3 0.0s
=> => sha256:148762f75a1f92cc9857e9c488bf95d5aac61e9905ec47a7408025b2dd5c3b7a 240B / 240B 1.7s
=> => sha256:ea1518237b3753b3fe40ee773d77651704178d9baa72ae5012e13a992cfa6c63 2.85MB / 2.85MB 2.9s
=> => sha256:4819c95424fc4a94767c9329b02238ebc0bc682384cb671379bc1fb8a12b55 10.94MB / 10.94MB 11.5s
=> => sha256:9875af95546db78168a6761b7fa205ed1cd0c153cd89356c1512e551c12b2d5c 622.29kB / 622.29kB 4.1s
=> => sha256:96526aa774ef0126ad0fe9e9a95764c5fc37f409ab9e97021e7b4775d82bf6fa 3.40MB / 3.40MB 6.3s
=> => extracting sha256:96526aa774ef0126ad0fe9e9a95764c5fc37f409ab9e97021e7b4775d82bf6fa 0.1s
=> => extracting sha256:9875af95546db78168a6761b7fa205ed1cd0c153cd89356c1512e551c12b2d5c 0.1s
=> => extracting sha256:4819c95424fc4a94767c9329b02238ebc0bc682384cb671379bc1fb8a12b55 0.2s
=> => extracting sha256:148762f75a1f92cc9857e9c488bf95d5aac61e9905ec47a7408025b2dd5c3b7a 0.0s
=> => extracting sha256:ea1518237b3753b3fe40ee773d77651704178d9baa72ae5012e13a992cfa6c63 0.1s
=> [internal] load build context 0.0s
=> => transferring context: 1.39kB 0.0s
=> [2/6] WORKDIR /code 0.1s
=> [3/6] RUN apk add --no-cache gcc musl-dev linux-headers 86.9s
=> [4/6] COPY requirements.txt requirements.txt 0.0s
=> [5/6] RUN pip install -r requirements.txt 10.6s
=> [6/6] COPY . . 0.0s
=> => exporting to image 5.6s
=> => exporting layers 4.5s
=> => exporting manifest sha256:7a2c5933236312efb6d3b3549f653c16ce30bb5206c33d44986c194948958d05 0.0s
=> => exporting config sha256:9b950c1bfc34b6a0dece7e281b18cb78762d2e501f10b9c5ba1c716f1f46acc2 0.0s
=> => exporting attestation manifest sha256:9688001898e3e79618f34fb894fb7c6dd5fe5754b9fee2548bac9712dad29fd 0.0s
=> => exporting manifest list sha256:4ca714e4801634fbb6f578413b227a65933e3c6cfa2e4c526c6c98169ff4582 0.0s
=> => naming to docker.io/library/docker-compose-web:latest 0.0s
[+] up 14/14king to docker.io/library/docker-compose-web:latest 1.0s
Image redis:alpine Pulled 42.5s
Image docker-compose-web Built 118.8s
Network docker-compose_default Created 0.1s
Container docker-compose-redis-1 Started 0.1s
Container docker-compose-web-1 Started 0.4s
[root@rhel9 docker-compose]#
```

```
[+] up 14/14king to docker.io/library/docker-compose-web:latest
✓ Image redis:alpine Pulled
✓ Image docker-compose-web Built
✓ Network docker-compose_default Created
✓ Container docker-compose-redis-1 Started
✓ Container docker-compose-web-1 Started
[root@rhel9 docker-compose]#
```

```
[root@rhel9 docker-compose]# docker compose up -d --build
WARN[0000] /root/docker-compose/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 9/10
  Image redis:alpine [=====] 36.7MB / 37.42MB Pulling 35.5s
[+] Building 3.2s (13/13) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 494B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 358B 0.0s
=> [internal] load metadata for docker.io/library/python:3.7-alpine 1.8s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/6] FROM docker.io/library/python:3.7-alpine@sha256:f3d31c8677d03f0b3c724446077f229a6ce9d3ac430f5c08cd7dff00292048c3 0.0s
=> => resolve docker.io/library/python:3.7-alpine@sha256:f3d31c8677d03f0b3c724446077f229a6ce9d3ac430f5c08cd7dff00292048c3 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 1.16kB 0.0s
=> CACHED [2/6] WORKDIR /code 0.0s
=> CACHED [3/6] RUN apk add --no-cache gcc musl-dev linux-headers 0.0s
=> CACHED [4/6] COPY requirements.txt requirements.txt 0.0s
=> CACHED [5/6] RUN pip install -r requirements.txt 0.0s
=> [6/6] COPY . . 0.1s
=> exporting to image 1.2s
=> => exporting layers 0.0s
=> => exporting manifest sha256:8a9848d056e09543d77d8b040cb1044cf25f999f99620b116b7f133ac4449e55 0.0s
=> => exporting config sha256:df9d3966cf128e6492c633a76fa1d254640963add66c66b9492b35f2e74d7b74 0.0s
=> => exporting attestation manifest sha256:ce6ebb4a823b11748b332c9ffa779b51669144f40abd29b1ba0eb986b8f62111 0.0s
=> => exporting manifest list sha256:c6488d727e54e88ed2ffaa796dad00da47a3ca6831114736a74ea0700d6cf106 0.0s
=> => naming to docker.io/library/docker-compose-web:latest 0.0s
[+] up 13/13king to docker.io/library/docker-compose-web:latest 1.1s
  Image redis:alpine Pulled 35.5s
  Image docker-compose-web Built 3.2s
  Container docker-compose-redis-1 Started 0.3s
  Container docker-compose-web-1 Started 0.4s
[root@rhel9 docker-compose]#
```

7. Go to browser <http://192.168.153.10:8000/>



Hello World! I have been seen 1 times.

curl http://192.168.153.10:8000

```
[root@rhel9 docker-compose]# curl http://192.168.153.10:8000/
Hello World! I have been seen 6 times.
[root@rhel9 docker-compose]#
```